



Impact analysis of climate data aggregation at different spatial scales on simulated net primary productivity for croplands



Matthias Kuhnert^{a,*}, Jagadeesh Yeluripati^{a,m}, Pete Smith^a, Holger Hoffmann^b, Marcel van Oijen^c, Julie Constantin^d, Elsa Coucheney^e, Rene Dechow^f, Henrik Eckersten^g, Thomas Gaier^b, Balász Grosz^f, Edwin Haas^h, Kurt-Christian Kersebaumⁱ, Ralf Kiese^h, Steffen Klatt^h, Elisabet Lewan^e, Claas Nendelⁱ, Helene Raynal^d, Carmen Sosa^e, Xenia Speckaⁱ, Edmar Teixeira^j, Enli Wang^k, Lutz Weihermüller^l, Gang Zhao^b, Zhigang Zhao^k, Stephen Ogleⁿ, Frank Ewert^b

^a Institute of Biological and Environmental Sciences, University of Aberdeen, 23 St. Machar Drive, Aberdeen AB24 3UU, UK

^b Crop Science Group, INRES, University of Bonn, Katzenburgweg 5, 53115 Bonn, Germany

^c Centre for Ecology and Hydrology, CEH-Edinburgh, Bush Estate, Penicuik EH26 0QB, UK

^d INRA, UMR 1248 AGIR & UR0875 MIA-T, F-31326 Auzeville, France

^e Department of Soil and Environment, Swedish University of Agricultural Sciences, Lennart Hjelm's väg 9, 750 07 Uppsala, Sweden

^f Thünen-Institute of Climate-Smart-Agriculture, Bundesallee 50, 38116 Braunschweig, Germany

^g Department of Crop Production Ecology, Swedish University of Agricultural Sciences, Ulls väg 16, 750 07 Uppsala, Sweden

^h Institute of Meteorology and Climate Research—Atmospheric Environmental Research, Karlsruhe Institute of Technology, Kreuzeckbahnstraße 19, 82467 Garmisch-Partenkirchen, Germany

ⁱ Institute of Landscape Systems Analysis, Leibniz Centre for Agricultural Landscape Research, 15374 Müncheberg, Germany

^j Systems Modelling Team (Sustainable Production Group), The New Zealand Institute for Plant and Food Research Limited, Canterbury Agriculture & Science Centre, Gerald St., Lincoln 7608, New Zealand

^k CSIRO Land and Water, Clunies Ross Street, Canberra, ACT, Australia

^l Institute of Bio- & Geosciences Agrosphere (IBG-3), Forschungszentrum Jülich, 52425 Jülich, Germany

^m The James Hutton Institute, Craigiebuckler, Aberdeen AB15 8 QH, UK

ⁿ Natural Resource Ecology Laboratory, Colorado State University, Fort Collins, CO 80523, USA

ARTICLE INFO

Article history:

Received 2 September 2015

Received in revised form 1 June 2016

Accepted 14 June 2016

Available online 29 June 2016

Keywords:

Net primary production

NPP

Scaling

Extreme events

Crop modelling

Climate

Data aggregation

ABSTRACT

For spatial crop and agro-systems modelling, there is often a discrepancy between the scale of measured driving data and the target resolution. Spatial data aggregation is often necessary, which can introduce additional uncertainty into the simulation results. Previous studies have shown that climate data aggregation has little effect on simulation of phenological stages, but effects on net primary production (NPP) might still be expected through changing the length of the growing season and the period of grain filling. This study investigates the impact of spatial climate data aggregation on NPP simulation results, applying eleven different models for the same study region ($\sim 34,000 \text{ km}^2$), situated in Western Germany. To isolate effects of climate, soil data and management were assumed to be constant over the entire study area and over the entire study period of 29 years. Two crops, winter wheat and silage maize, were tested as monocultures. Compared to the impact of climate data aggregation on yield, the effect on NPP is in a similar range, but is slightly lower, with only small impacts on averages over the entire simulation period and study region. Maximum differences between the five scales in the range of 1–100 km grid cells show changes of 0.4–7.8% and 0.0–4.8% for wheat and maize, respectively, whereas the simulated potential NPP averages of the models show a wide range ($1.9\text{--}4.2 \text{ g C m}^{-2} \text{ d}^{-1}$ and $2.7\text{--}6.1 \text{ g C m}^{-2} \text{ d}^{-1}$ for wheat and maize, respectively). The impact of the spatial aggregation was also tested for shorter time periods, to see if impacts over shorter periods attenuate over longer periods. The results show larger impacts for single years (up to 9.4% for wheat and up to 13.6% for maize). An analysis of extreme weather conditions shows an aggregation effect in vulnerability up to 12.8% and 15.5% between the different resolutions for wheat and maize, respectively. Simulations of NPP averages over larger areas (e.g. regional scale) and longer time periods (several years) are relatively insensitive to climate data

* Corresponding author.

E-mail address: matthias.kuhnert@abdn.ac.uk (M. Kuhnert).

aggregation. However, the scale of climate data is more relevant for impacts on annual averages of NPP or if the period is strongly affected or dominated by drought stress. There should be an awareness of the greater uncertainty for the NPP values in these situations if data are not available at high resolution. On the other hand, the results suggest that there is no need to simulate at high resolution for long term regional NPP averages based on the simplified assumptions (soil and management constant in time and space) used in this study.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Net primary production (NPP) is a crucial variable characterising the condition of an ecosystem (Pan et al., 2014) and it is sensitive to climate change. Spatial NPP is difficult to measure and is often biased and uncertain (Pan et al., 2014), because measurements show several limitations (indirect determination, spatially and temporally limited). Spatial modelling is an important tool for interpolation and extrapolation of measurements or for providing spatially distributed projections at regional (Reich et al., 1999; Zaehle et al., 2006; Bandaru et al., 2013; Liu et al., 2015), continental (Ciais et al., 2010) and global scales (Hemming et al., 2013; Friend et al., 2014). The regional scale is relevant for policy makers to analyse adaptation and mitigation strategies, but NPP data at this scale are often derived by extrapolating measured information from the site scale to a region by applying models developed at site scale (Zhang et al., 2015). This model-based up-scaling requires a balance between accuracy and simulation time.

Spatial modelling of NPP relies on spatially distributed input and driving data like weather data and information on soil, land use and management characteristics. Depending on environmental parameters, ecosystem characteristics and the chosen resolution, the impacts of extrapolation or interpolation may be great or small since there is a higher uncertainty for high relief areas compared to relatively flat areas, as shown by Pierce and Running (1995). For this reason, estimates of error and uncertainty arising from data aggregation across scales needs to be quantified.

Several studies have highlighted the impact of data aggregation on simulation results (Cale et al., 1983; Rastetter et al., 1992; Ewert et al., 2015; Zhao et al., 2015). de Wit et al. (2005) and Hoffmann et al. (2015) investigated the impact of climate data aggregation on crop yields. While de Wit et al. (2005) varied precipitation and solar radiation only on the resolutions 10 km and 50 km, Hoffmann et al. (2015) differentiated between five different resolutions between 1 and 100 km and also considered aggregation effects of temperature for 13 models. Both studies found only slight impacts of data aggregation on simulated yield over longer time periods at a regional scale. van Bussel et al. (2011) investigated the impacts of climate aggregation on croplands and focused on phenological stages rather than primary production, but they also found minor effects on simulated average values. The impacts of climate data aggregation on NPP were tested by Nungesser et al. (1999) and Pierce and Running (1995), for American forests. In both studies, the impact was minor for averages over the entire study area, but showed impacts for smaller areas, especially for areas dominated by strong relief changes (Pierce and Running, 1995). In both studies, the effects were tested by one model and for two resolutions of 10 and 50 km grid cells in Nungesser et al. (1999), and 1 km and 110 km in Pierce and Running (1995). The latter study investigated the effect for different input variables (relief, climate and soil) and found that climate data aggregation was the dominant variable affecting scale differences of NPP. They also observed larger scale effects for shorter time periods, which could be an indication of extreme weather events that average out over larger areas or longer time periods. Overall, regional simulation results over longer peri-

ods seem to be little affected by climate data aggregation. Over longer periods, changes of NPP even out and the impact of extreme events may not be obvious in a long term average, but relevant for shorter periods. Reichstein et al. (2013) describe the importance of temporal and spatial scale for detecting impacts of extreme weather conditions on the carbon balance, and highlight the risk of missing extreme weather conditions by integration of weather data across scales. Impacts of extreme weather also depend on the temporal scale, which has not so far been tested for impacts on NPP. However, studies determining the effect on NPP simulations of croplands are rare, and no study to date has analysed the potential to miss extreme events with climate data aggregation.

The objective of this paper, therefore, is to quantify error and uncertainty of NPP simulations of croplands caused by climate data aggregation across five resolutions (1, 10, 25, 50 and 100 km grid cell side length). This study addresses the three questions: (i) what are the impacts on long term NPP averages over the entire region?, (ii) how does the aggregation effect change over shorter time periods?, (iii) is the aggregation effect more pronounced in years with extreme weather conditions compared to “normal” years? These questions are answered by using a simulation approach involving eleven different models. Additionally, a vulnerability analysis helps to identify the impact of climate data aggregation for years with extreme weather conditions. Thus, we provide the first systematic analysis considering the impact of spatial weather data aggregation on NPP using five resolutions and 11 different models.

2. Methods

2.1. Aggregation effect

Crop models are generally designed and parameterised for a specific spatial resolution, which can be for points, plots, fields, map grid cells, or larger areas. When a model is run at a resolution different from that for which it was designed, inputs must be spatially aggregated or dis-aggregated, which will cause errors in outputs if the model is nonlinear. Most crop models are complex nonlinear point models. When such models are used to evaluate yield potential in large regions, lack of input information and computational constraints means that they need to be run at coarser resolution. The ensuing aggregation error is difficult to predict because of model complexity and our limited knowledge of the spatial heterogeneity underlying our aggregated input values. Input aggregation thus adds to the uncertainty associated with model predictions (Heuvelink and Pebesma, 1999). One way of reducing uncertainty about input aggregation error is to evaluate models at different resolutions for areas where high-resolution input data are available. In such cases, we are able to block-average the inputs and quantify the effect that the aggregation has on model outputs. Building up experience with such case-studies may help us assign probabilities to different degrees of aggregation error in comparable areas. In this study, we quantify the effect of climate data aggregation on the simulation of NPP, and define the aggregation effect ($E_{\text{aggregation}}$)

as:

$$E_{\text{aggregation}} = \frac{\max(\text{VAL}_{\text{res}_1}, \dots, \text{VAL}_{\text{res}_n}) - \min(\text{VAL}_{\text{res}_1}, \dots, \text{VAL}_{\text{res}_n})}{\text{VAL}_{\text{res}_1}} \quad (1)$$

Based on Eq. (1) the aggregation effect is defined as the maximum difference between the simulated NPP averages at the different resolutions, and is quantified by the difference between maximum and minimum NPP averages of the different resolutions (1, 10, 25, 50 and 100 km):

$$E_{\text{aggregation,model}} = \frac{\max(NPP_{\text{Res1,model}}, \dots, NPP_{\text{Res100,model}}) - \min(NPP_{\text{Res1,model}}, \dots, NPP_{\text{Res100,model}})}{NPP_{\text{Res1,model}}} \quad (2)$$

with $NPP_{\text{Res1,model}}$ as the average simulated NPP for one model at the resolution 1 km. This approach allows a model-specific calculation of the aggregation effect, and is independent of any trends towards the coarser resolution. The difference describes the maximum expected bias by picking one resolution and comparing to the model results at any other resolution. Therefore, this method is applicable to different spatial or temporal averages.

The aggregation effect can also be assessed based on ensemble runs. $E_{\text{aggregation}}$ can be used to calculate the resolution-specific averages over all models with the formulation:

$$E_{\text{aggregation,average}} = \frac{\max(NPP_{\text{Res1}}, \dots, NPP_{\text{Res100}}) - \min(NPP_{\text{Res1}}, \dots, NPP_{\text{Res100}})}{NPP_{\text{Res1}}} \quad (3)$$

with the averages of NPP representing the averages of all cells for all models of the considered resolution.

2.2. Study area

The study area is the state North Rhine-Westphalia situated in the West of Germany with a temperate humid climate influenced by the proximity of the Atlantic Ocean. The state is 34084 km² in size with an elevation from 0 to 827 m above sea level, with lower plains in the North West and higher elevations in the South-East. The land-use is dominated by agriculture (more than 60% of the area), but in this study the entire area (including the 40% of forest, urban areas and infrastructure as well as water bodies) was considered to be cropland. To standardize the simulation runs monocultures of either winter wheat (*Triticum aestivum* L.) or silage maize (*Zea mays* L.) were assumed for the entire area.

The daily weather data are provided at five different resolutions (1, 10, 25, 50 and 100 km grid cells), while soil data (typical soil type) and management (good agricultural practice) were assumed to be constant during the study period and over the entire study area. The chosen soil type is a sandy loam, which is typical for this region and the management includes ploughing, sowing, harvest and three (130, 52, 26 kg ha⁻¹) and two (30, 208 kg ha⁻¹) fertilizer applications during spring for wheat and maize, respectively (for details see Hoffmann et al., 2015). The models were not calibrated for the study area, but were adjusted based on 30-year yield averages (1982–2011) of about 8 t ha⁻¹ for winter wheat and 14 t ha⁻¹ for silage maize. The weather data, presented and discussed by Zhao et al. (2015) and Hoffmann et al. (2015), show a 30-year average temperature of 9.7 °C, an average annual precipitation of 899 mm and mean annual global radiation of 3758 MJ m⁻² a⁻¹ (1982–2011) with the standard deviations of 1.2 °C, 214.0 mm a⁻¹ and 169.4 MJ m⁻² a⁻¹, respectively. The coldest year was 2004, with an average temperature of 8.9 °C and the warmest year was 1983 (11.2 °C). The driest year was 2001 with precipitation of only 516.2 mm. All coarser resolutions of the weather data were based on the grid cells of the 1 km resolution for daily time steps. The data show a decrease of temperature (from 9.7 °C to 9.4 °C) and precipi-

tation (from 899 mm to 824 mm) starting from the 1 km resolution to the coarsest resolution of 100 km.

2.3. Modelling applications

There are three different approaches using different model settings to analyse the impact of different processes contributing to the simulation of NPP. In a first approach, no limitations to growth factors, other than temperature and radiation, are simulated explicitly (switched off or compensated in all models). We denote this

potential, non-limited growth, potential NPP or PN. The second approach considers only water-limitation (WN), while the third approach considers nitrogen and water limitation (NN). The way limitations are switched off differs between the models. Some models switched off the stress factors, other models compensated the stress by providing additional water and nutrient applications.

The settings for management are presented by Hoffmann et al. (2015). The sowing date is fixed for all models, but the harvest date can vary for the phenological models. For all models the date

for the latest harvest is fixed. If the phenological model does not determine maturity before this date, there will be an automatic harvest regardless of the model status. Models without internal phenological models used this date as harvest date.

2.4. Models

Eleven models were used in this study; eight crop models and three biogeochemical models (Table 1). All models require data on a daily time step (except CENTURY which uses a monthly time step), consider the complete range of management practices (except AgroC that does not consider nitrogen limitation) and provide simulations for the two considered crops, wheat and maize (except

Table 1
List of the participating models.

No.	Model	Model version	References
1	HERMES	4.26	Kersebaum (2007, 2011)
2	APSIM	7.5	Keating et al. (2003), Holzworth et al. (2014)
3	COUP	5	Conrad and Fohrer (2009), Jansson and Karlberg (2004)
4	DailyDayCent	from 1st of May 2013	Del Grosso et al. (2001, 2006), Parton et al. (2001), Yeluripati et al. (2009)
5	LandscapeDNDC	0.21	Haas et al. (2012), Kraus et al. (2015)
6	LINTUL	SIMPLACE <LINTUL5;SLIM>	Van Ittersum et al. (2003), Shibu et al. (2010), Gaiser et al. (2013)
7	MONICA	1.2	Nendel et al. (2011)
8	STICS	8.3	Bergez et al. (2013), Brisson et al. (1998, 2008)
9	APSIMmod	7.5 ^a	Chen et al. (2010), Keating et al. (2003), Wang et al. (2002)
10	CENTURY	4.6	Parton et al. (1993, 1995)
11	AgroC	1.0	Herbst et al. (2008)

^a The model version is modified (new temperature function for phenology and biomass growth for wheat crop).

COUP that only simulates wheat). The growing season for the crop models is determined by internal phenological models based on a fixed sowing date, while a fixed harvest date is set for the three biogeochemical models CENTURY, DailyDayCent and LandscapeDNDC, and the crop model STICS (i.e. fixed length of the growing season).

Five of the models determine NPP based on the radiation use efficiency concept (AgroC, APSIM, APSIMmod, COUP, LINTUL, STICS), while other models determine NPP based on the difference between gross primary production and respiration (HERMES, MONICA), calculated directly (DailyDayCent) or other approaches (LandscapeDNDC). More details about the models are provided in Hoffmann et al. (2015).

2.5. Evaluation of aggregation effects over different time periods

The simulation results (NPP averages over the entire study area) were averaged over different periods (1, 5, 10, 15, 20, 25, 29 years) to examine the maximum differences between the five resolutions, as influenced by the different temporal scales. The number of averages considered varies for the different time periods (29, 6, 3, 2, 2, 2, 1, respectively). The analysis for the periods 20 and 25 years were applied twice, covering mainly the first and the last years with some data overlap. The results are presented as the mean aggregation effect as well as the maximum aggregation effect between the five resolutions.

2.6. Vulnerability analysis

Vulnerability and risk are terms that are widely used in different communities and described in different contexts with different definitions. In this study we use an approach developed by Van Oijen et al. (2014), designed to investigate impacts of extreme weather events on carbon dynamics. The approach, based on an abiotic definition of extreme periods, compares the impacts on a chosen biotic ecosystem variable of the defined “extreme” or “hazardous” and “not extreme” or “non-hazardous” conditions. Van Oijen et al. (2014) chose the standardized precipitation evapotranspiration index (SPEI), developed by Vicente-Serrano et al. (2010), as the abiotic factor separating hazardous from non-hazardous conditions. SPEI is a drought index based on the difference between potential evapotranspiration and precipitation. There is no fixed threshold to define an extreme drought impact or growth reducing conditions and SPEI can be calculated for any duration. Additionally, the index is normalized and normally distributed. The average is about 0 for the considered period of 1982–2011 with 64% of the values between -1 and $+1$ and 19% below -1 at 1 km resolution. These statistics stay the same for all resolutions, with the excep-

tion of the number of values in the -1 to $+1$ interval, which drops to 63% for the 100 km resolution. The potential evapotranspiration can be calculated with different approaches; in this study the method developed by Thornthwaite (1948) was used.

SPEI is one of the indices that considers both, precipitation and temperature in the calculation, rather than only precipitation, but is still easy to apply, which makes it an attractive index to use in this study. Van Oijen et al. (2014) suggest two thresholds to separate hazardous from non-hazardous conditions: $\text{SPEI} < -1$ and $\text{SPEI} < -2$. For the actual study region there is only a small number of SPEI values below -2 , so $\text{SPEI} < -1$ was chosen as the threshold. Following Van Oijen et al. (2014), the period to calculate the SPEI is restricted to half a year. In contrast to the approach of Van Oijen et al. (2014), who suggested the period April–September, the period February–July was used in this study to better reflect the crop growth period. The system variable used in this study is NPP.

Vulnerability (V) describes a possible damage/impact on a system and the risk (R) is described by the product of the probability (P) that a hazardous event (H) occurs and its impact on the system.

$$R = P(H) \times V \quad (2)$$

This equation represents the relation between probability, and vulnerability and can be expressed by using the reduction of the NPP by the hazardous periods, described as risk:

$$R = E(\text{NPP}|\text{non-hazardous}) - E(\text{NPP}) \quad (3)$$

$E(\text{NPP}|\text{non-hazardous})$ is the average value of NPP for all grid cells and years with a $\text{SPEI} \geq -1$ and $E(\text{NPP})$ is the overall average of NPP (including both, hazardous and non-hazardous conditions). The vulnerability describes the difference of the NPP averages for the non-hazardous and the hazardous years and grid cells.

$$V = E(\text{NPP}|\text{non-hazardous}) - E(\text{NPP}|\text{hazardous}) \quad (4)$$

3. Results

3.1. NPP differences between the models

The different model simulations of NPP are compared for the 1 km grid resolution, which is considered to be the most accurate simulation for each model, and is thus used as the baseline. The results for wheat vary for the different models with mean values for 29 years of $2.0\text{--}3.9 \text{ g C m}^{-2} \text{ d}^{-1}$, $2.0\text{--}3.8 \text{ g C m}^{-2} \text{ d}^{-1}$ and $1.9\text{--}3.8 \text{ g C m}^{-2} \text{ d}^{-1}$ NPP for PN, WN and NN, respectively (Fig. 1). The simulation results of STICS and AgroC show the highest variation over the 29 years, while the results of COUP and LandscapeDNDC vary only within a narrow range. Some of the models are sensitive to water limitation (e.g. COUP), which is reflected by

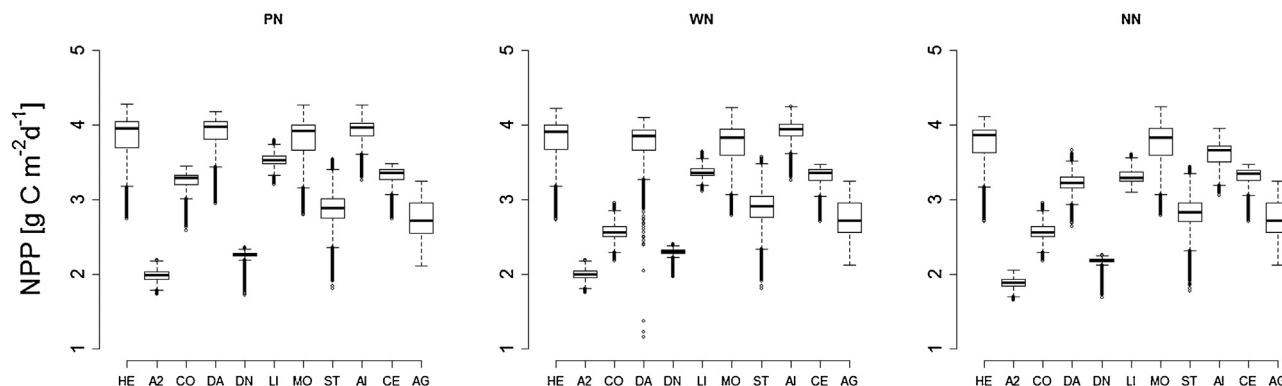


Fig. 1. Simulated NPP for winter wheat at 1 km resolution for potential growth (PN), under water limitation (WN) and under nutrient and water limitation (NN). The models are in the order HERMES (HE), APSIMmod (A2), COUP (CO), DailyDayCent (DA), LandscapeDNDC (DN), LINTUL (LI), MONICA (MO), STICS (ST), APSIM (AI), CENTURY (CE) and AgroC (AG).

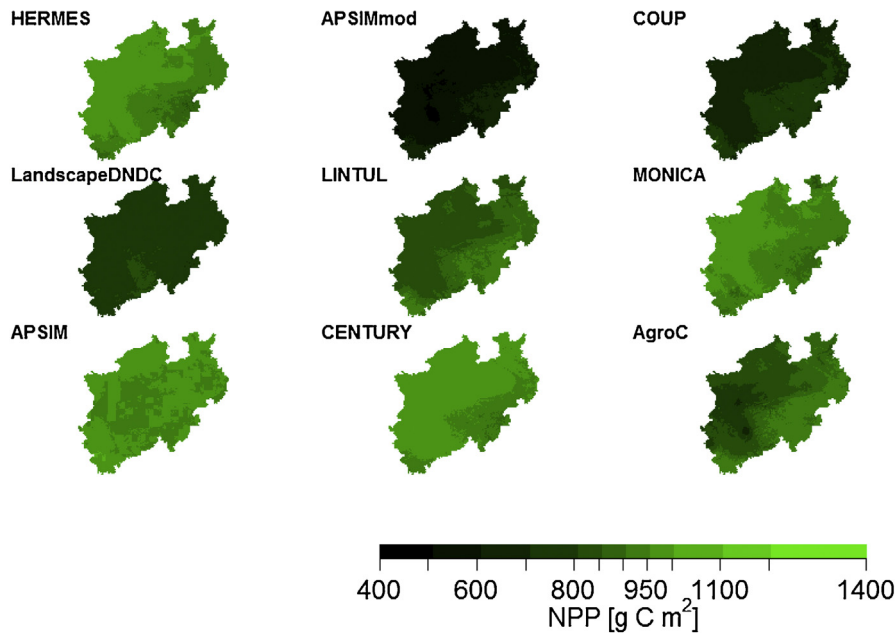


Fig. 2. Spatial distribution of the 29 year averages of NPP over the growing season for the 11 models assuming wheat mono-culture for the PN approach.

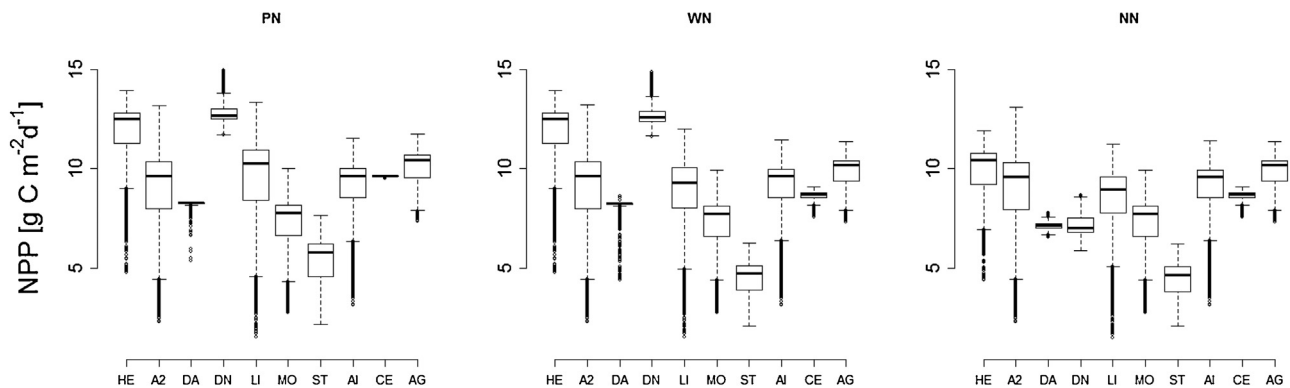


Fig. 3. Simulated NPP values for silage maize at 1 km resolution for potential growth (PN), under water limitation (WN) and under nutrient and water limitation (NN). The models are in the order HERMES (HE), APSIMmod (A2), DailyDayCent (DA), LandscapeDNDC (DN), LINTUL (LI), MONICA (MO), STICS (ST), APSIM (AI), CENTURY (CE) and AgroC (AG).

the differences between the PN and the WN approach (Fig. 1), other models show minimal impacts of water limitation (e.g. STICS). The spatial distribution of the results shows a dependency on elevation (Fig. 2). However, the spatial distribution of high and low NPP values is different between the models. While most models show higher NPP values for low elevation and lower values for higher elevations, another group of models (AgroC, COUP, LINTUL, APSIM and APSIMmod) show the opposite spatial separation of high and low NPP values.

The simulated NPP for maize ($7.4\text{--}12.8\text{ g C m}^{-2}\text{ d}^{-1}$, $7.3\text{--}12.7\text{ g C m}^{-2}\text{ d}^{-1}$ and $7.1\text{--}10.0\text{ g C m}^{-2}\text{ d}^{-1}$ for PN, WN and NN, respectively) show higher maximum values than the NPP for wheat, and indicate an even lower sensitivity to water limitation, which is represented by a comparison of the simulation results of PN and WN (Fig. 3). The extreme NPP values for APSIMmod and LandscapeDNDC for wheat and maize, respectively, are outside the range of the other models, but are included in all analyses.

3.2. Model specific aggregation effect

The aggregation effect as described in Eq. (1) shows a range for all models of 0.4–7.8% and 0–4.8% for wheat and maize, respectively

(Table 2). The analysis of the medians shows slightly larger aggregation effects with 0.3–11.4% and 0.0–10.0% for wheat and maize, respectively (Table 2). There are no obvious trends in the changes of NPP from the 1 km resolution to 100 km resolution, neither for the crops nor for the different models (Figs. 4 and 5). However, the models LandscapeDNDC, MONICA, CENTURY and DailyDayCent show relatively small changes (<1.2%) for both crops, and HERMES for the wheat simulations, while APSIM, APSIMmod and AgroC show relatively high aggregation effects (more than 4.8%) between the different scales. The aggregation effect varies between the models as does the trend. While APSIMmod, LINTUL, AgroC and APSIM show increasing NPP values towards coarser resolutions for the wheat simulations, HERMES, MONICA, CENTURY and DailyDayCent show decreasing NPP. The simulation results of COUP and STICS show no trend, but have minimum NPP averages for the resolutions of 10 km and 50 km, respectively. The median is affected more than the average values for some models, especially for the maize simulations, and most models show stronger changes for WN and NN than for PN (Table 2). The results for maize support the findings of the wheat simulations, but the scale effect is smaller, and effects and trends differ for some models between the two crops. HERMES and DailyDayCent show minimal differences between the resolu-

Table 2
Relative maximum differences of NPP averages (AVG) and median (MED) between the five resolutions [%]. The values represent the simulation results of wheat (W) and maize (M) for the three approaches (PN, WN, NN). All differences are related to the resolution with the lowest NPP average. The models are HERMES (HE), APSIMmod (A2), COUP (CO), DailyDayCent (DA), LandscapedNDC (DN), LINTUL (LI), MONICA (MO), STICS (ST), APSIM (AI), CENTURY (CE) and AgroC (AG). Changes of >3% are highlighted by grey boxes.

	HE	A2	CO	DA	DN	LI	MO	ST	AI	CE	AG
AVG W PN	0.9	5.0	2.2	1.1	n.s.	4.7	0.8	2.0	4.9	0.9	7.8
AVG W WN	0.8	4.8	2.2	0.9	0.2	2.4	1.1	1.8	4.8	0.8	7.8
AVG W NN	0.5	5.4	2.2	0.7	1.6	3.0	1.2	2.0	5.4	0.7	n.s.
AVG M PN	1.3	2.0	n.s.	0.5	1.3	2.0	2.0	1.5	2.0	0.0	3.2
AVG M WN	1.3	2.0	n.s.	0.2	1.2	2.5	1.5	2.0	2.0	0.5	3.5
AVG M NN	4.8	2.0	n.s.	1.1	2.8	1.7	1.5	2.0	2.0	0.6	n.s.
MED W PN	0.9	4.9	2.2	2.4	n.s.	4.7	0.8	2.0	4.9	0.9	7.8
MED W WN	1.5	7.8	2.3	2.2	0.3	3.2	2.9	2.1	7.8	1.4	11.4
MED W NN	0.9	8.8	2.3	1.9	1.0	3.8	3.0	2.3	8.8	1.3	n.s.
MED M PN	0.7	10.0	n.s.	0.2	2.8	8.7	3.8	3.3	10.0	0.0	3.1
MED M WN	0.7	10.0	n.s.	0.2	2.7	10.0	3.3	3.7	10.0	0.3	3.9
MED M NN	4.0	10.0	n.s.	0.6	5.5	8.8	3.3	3.8	10.0	0.3	n.s.

n.s. not simulated.

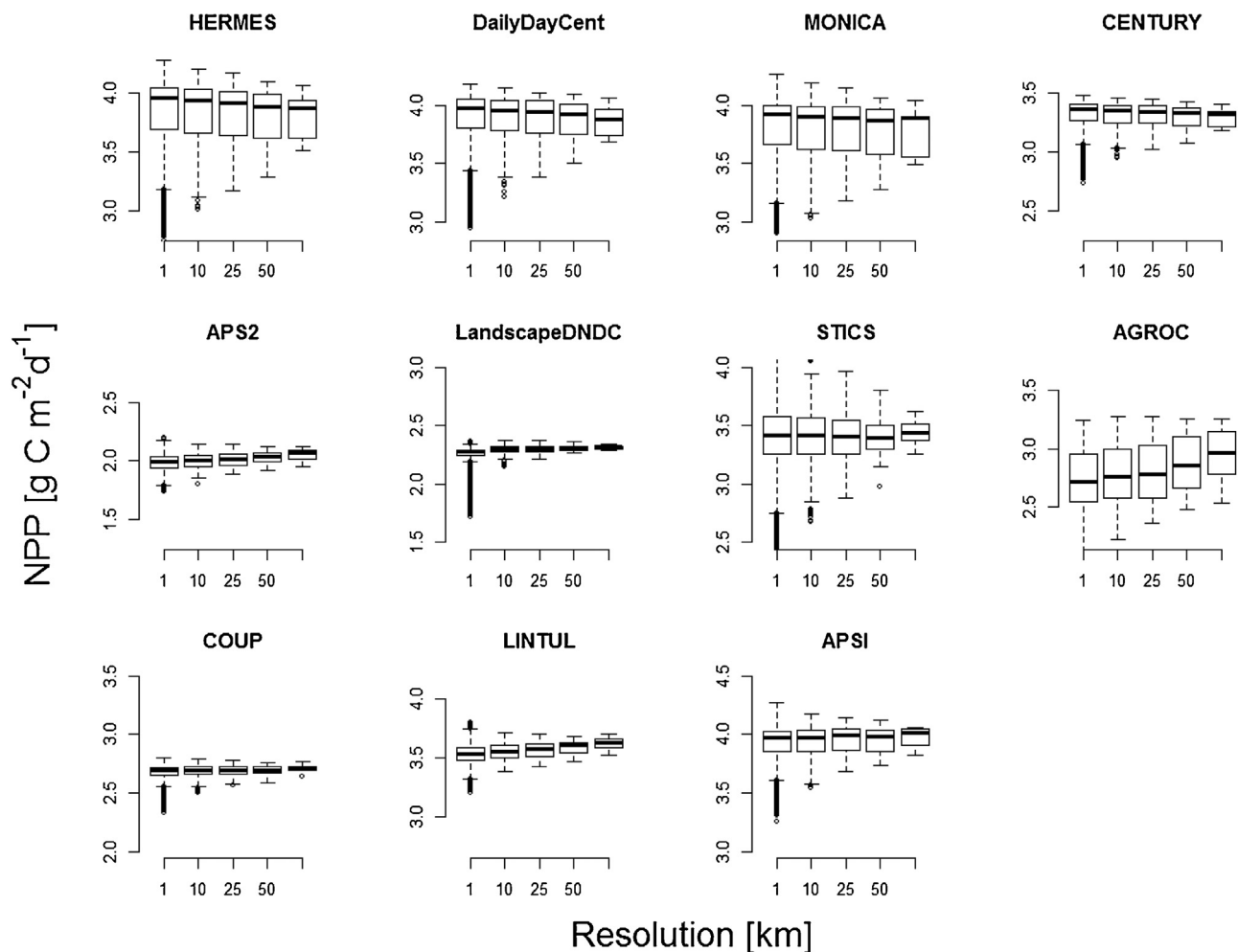


Fig. 4. Simulated NPP of wheat for the potential growth (PN). These boxplots represent the variability over 29 NPP averages over the growing season for the five resolutions (1, 10, 25, 50 and 100 km).

tions, while APSIMmod, LINTUL, MONICA, STICS and APSIM show a decreasing trend, with AgroC and LandscapedNDC showing an increasing trend towards coarser resolutions.

The aggregation effect for the model ensemble is calculated for the resolution specific average over all models (Eq. (3)). The effect is below 0.9% for wheat and 2.0% for maize, which is smaller than all aggregation effects of the individual models.

3.3. Aggregation effect over different time periods

The impact of scales is also tested for periods shorter than 29 years (Figs. 6 and 7). The simulation results are averaged for each time step (according to the considered period of 1, 5, 10, 15, 20, 25 years) over the entire area (for each resolution separately). While the maximum aggregation effect is strongest for a single year and does not change greatly for periods of 10 years or longer, the aver-

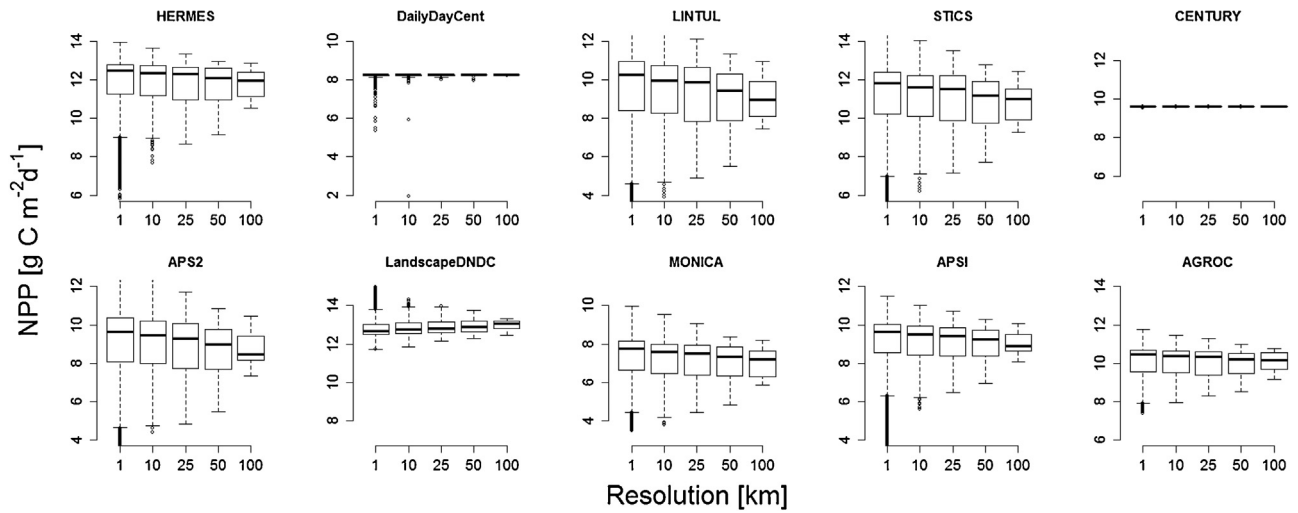


Fig. 5. Simulated NPP of maize for the potential growth (PN). These boxplots represent the variability over 29 NPP averages over the growing season for the five resolutions (1, 10, 25, 50 and 100 km).

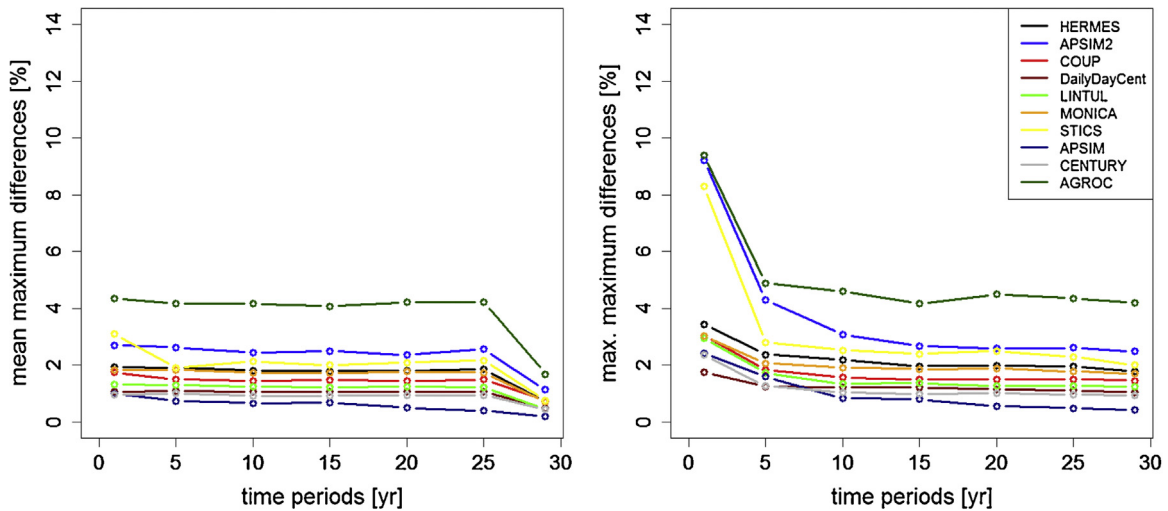


Fig. 6. The relative differences between the maximum and minimum NPP (wheat PN) between the resolutions for data averaged over different time periods (annual to 29 year averages). On the left side the differences are averaged for each period, while the right side shows the maximum values for each period.

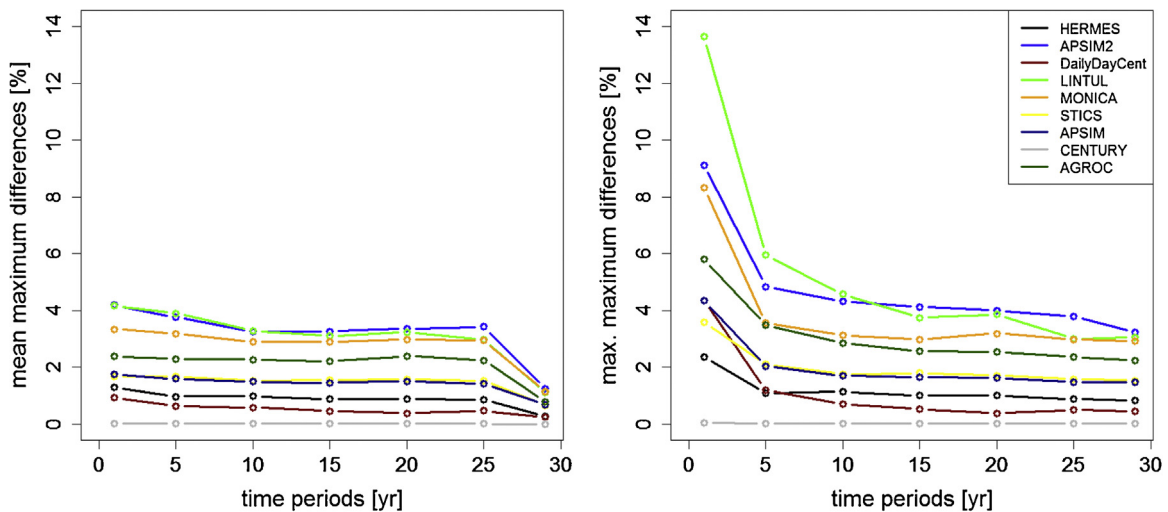


Fig. 7. The relative differences between the maximum and minimum NPP (maize PN) between the resolutions for data averaged over different time periods (annual to 29 year averages). On the left side the differences are averaged for each period, while the right side shows the maximum values for each period.

age aggregation effect stays almost the same, but decreases for the 29 year period. The effects are the same for all models for both crops (Figs. 6 and 7), but the values differ.

The NPP values in this study represent NPP during growing season. Length of growing season varies between the different models and the different years (Table 3), because of different phenological sub-models and inter-annual variations of temperature. As already mentioned, the date for the latest possible harvest is fixed and this date is used as actual harvest by the models CENTURY, DailyDayCent and LandscapeDNDC. The greatest differences between the lengths of the growing season are observed for LINTUL and AgroC, while there are relatively small changes for HERMES and MONICA.

3.4. Vulnerability analysis

The results of the vulnerability analysis are represented in Figs. 8 and 9, showing vulnerability (grey bars) and the risk (black bars) for each model and resolution. Negative values indicate a higher NPP average for hazardous conditions than for the non-hazardous conditions (vulnerability) or a higher NPP for the overall average than for the non-hazardous conditions (risk). The analysis shows positive values for vulnerability and risk for all simulation results of wheat, except for CENTURY, AGROC and the two APSIM models (Fig. 8). The values between the different resolutions vary for the different models. The vulnerability analysis for the maize simulations shows a negative risk and vulnerability for APSIMmod, CENTURY and LandscapeDNDC, while LINTUL, APSIM and DailyDayCent vary between the resolutions (Fig. 9).

Overall, vulnerability and risk differ for most models depending on the resolution, but there is no clear trend for increase or decrease of vulnerability or risk towards coarser resolution. The average risk for wheat simulations is about $0.02 \text{ g C m}^{-2} \text{ d}^{-1} \pm 0.02 \text{ g C m}^{-2} \text{ d}^{-1}$, with no trend between the different resolutions and the average vulnerability of $0.13 \text{ g C m}^{-2} \text{ d}^{-1}$ with a standard deviation of $0.10 \text{ g C m}^{-2} \text{ d}^{-1}$ shows also no clear trend. The differences of

Table 3

Minimum and maximum average length of the growing season of wheat for each resolution (overall averages) as determined in the different models. The shortest length represents, in all cases, the length of the growing season for resolution of 1 km and longest growing season was for all models the resolution of 100 km. The maximum length of the growing season is used for models without and internal calculation. This maximum length is 304 days and 150 days for wheat and maize, respectively.

model	min length [d]	max length [d]	change [d]
HERMES	256.1	257.6	1.5
APSIMmod	249.9	253.9	4.0
COUP	239.3	241.7	2.4
LINTUL	243.9	249.1	5.2
MONICA	259.5	261.1	1.6
STICS	236.4	239.0	2.6
APSIM	254.7	257.1	2.4
AgroC	238.2	243.3	5.1

vulnerability between the five resolutions show the maximum difference between 1.6% for CENTURY and 12.8% for LINTUL (maximum difference relative to the average NPP of resolution of 1 km) for wheat and between 1.1% (CENTURY) to 15.5% (MONICA) for maize. In these calculations the models AgroC, APSIM, APSIMmod and DailyDayCent for wheat, and APSIMmod, LandscapeDNDC and LINTUL for maize are not considered, because the results of these models indicate no vulnerability to drought under these conditions. In contrast to wheat, the vulnerability analysis of maize shows mainly negative values (Fig. 9), except for STICS and AgroC (positive values), and DailyDayCent and LINTUL (varying values). The number of values (cells and years with a SPEI < -1) may affect the results, but the number of extreme cells (based on SPEI) is within a narrow range of 17.6–18.2% for the different resolutions, so the relative numbers of hazardous cells stays about the same.

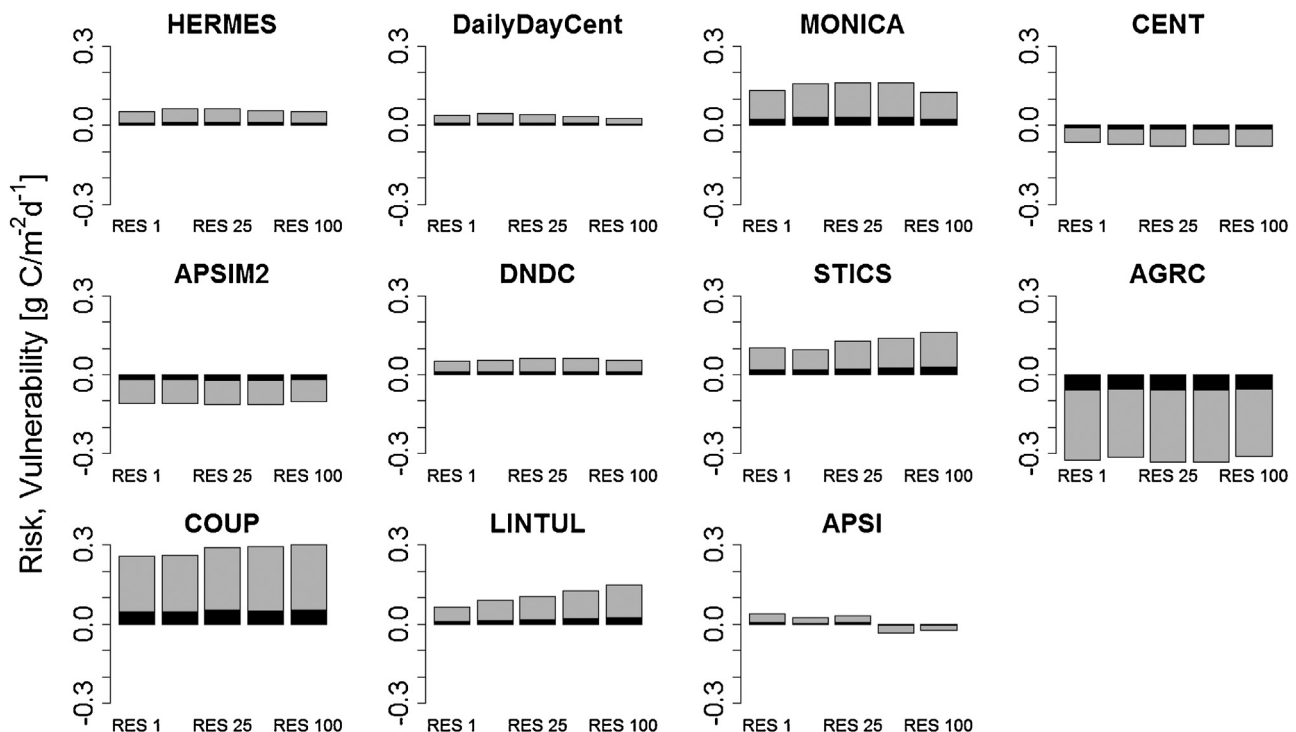


Fig. 8. Vulnerability (grey bar) and risk (black bar) for 11 models for water limitation simulation results of NPP (WN). The results represent the simulation results for wheat for the period 1983–2011. The terms vulnerability and risk are used according to the definition by van Oijen et al. (2014) and describe the impacts of hazardous in comparison to non-hazardous conditions (see also section 2.6).

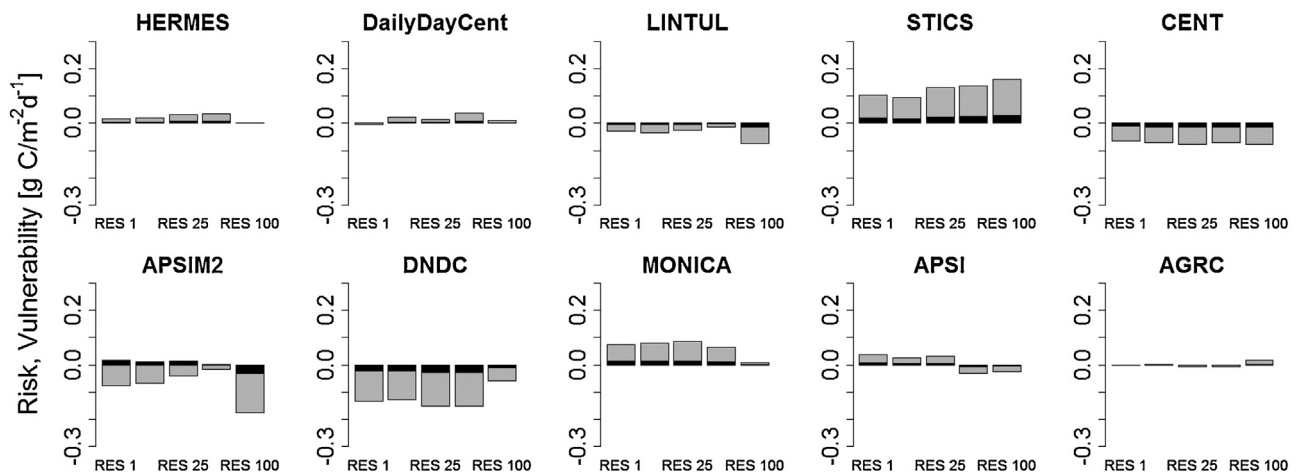


Fig. 9. Vulnerability (grey bar) and risk (black bar) for 10 models simulated for water limitation considered (WN). The results represent the simulation results for maize for the period 1983–2011. The terms vulnerability and risk are used according to the definition by van Oijen et al. (2014) and describe the impacts of hazardous in comparison to non-hazardous conditions (see also section 2.6).

4. Discussion

4.1. NPP differences between the models

The simulated potential NPP averages for the growing season for the 1 km resolution simulations range from 1.8 to 4.1 $\text{g C m}^{-2} \text{d}^{-1}$ and 7.4–12.8 $\text{g C m}^{-2} \text{d}^{-1}$ for wheat and maize, respectively, which is higher than the reported annual NPP averages of European croplands of $550 \pm 50 \text{ g C m}^{-2} \text{yr}^{-1}$ (Schulze et al., 2010). This is expected for these crops in the study region due to favourable climatic and soil conditions. Because the 1 km grid maps are the highest resolution, we assume these data as the most accurate and, therefore, they were used as baseline, because detailed spatial measurements of crop yields are not available. As mentioned, the results are based on simulation runs of uncalibrated models, but adjusted to proxies for a 30 year average of crop yield. In general, for spatial modelling, as performed here, data for calibration are rarely available or, if available, often restricted in space to one or a few point measurements. This makes appropriate calibration for spatial modelling difficult, and the adjustment to a 30 year proxy is an alternative to set up the model and capture NPP within a reasonable range. The two models with low NPP for wheat (APSIMmod) and high NPP for maize (LandscapeDNDC) are most likely under- and over-estimates of NPP, because of the lack of calibration. As the results are not unrealistic for crop yields in central Europe, the results from both models are still included in our analysis, because the bias affects all resolutions. NPP differs between the models up to 2.3 and 5.4 $\text{g C m}^{-2} \text{d}^{-1}$ for wheat and maize, respectively. Yield, which is the target variable for the adjustment of the models, ranges from 7.6 to 8.7 and 15.4–17.6 t ha^{-1} for wheat and maize, respectively, which has a smaller discrepancy between the models (Hoffmann et al., 2015). As mentioned above, the approaches for calculating NPP in the models are conceptually different, and by grouping the models according to the approach used to estimate radiation use efficiency, differences between GPP and respiration or direct calculation of NPP reduced the differences within the groups. The two exceptions for wheat (APSIMmod) and for maize (LandscapeDNDC) are already mentioned above. This shows that model structure affects the NPP and the differences in the structure induce the wide range of NPP averages.

A comparison between the PN, WN and NN approaches enables water limitation and nitrogen limitation impacts on NPP to be detected. Nitrogen limitations play a minor role for the study

region, because of sufficient fertilization. As Fig. 1 shows, the results of the COUP model indicate a strong sensitivity to water limitation (26–30% decrease of average NPP), and LINTUL is also sensitive to water limitation (4.7–6.1% decrease of average NPP), while the other models only show modest sensitivity to drought stress on the average (all <2% difference). The differences in sensitivity between the models do not show a specific impact on the aggregation affect. Neither the strength of the effect nor the changes between PN and PW are similar to each other or different to the other models.

The contradictory spatial distribution of high and low NPP values reflect the diversity of crop parameters and phenological sub-models applied in the different models. In contrast to the NPP, the distribution of yield does not necessarily show a similar spatial pattern in the study area. These differences between NPP and yield are related to the different impacts of temperature changes on the simulation of phenological stages, which affects the lengths of the growing season differently to the length of the grain filling period. There is additional biomass production of wheat from the extension of the growing season, while the grain filling period does not necessarily benefit from warmer climatic conditions. This is reflected in the results for the year 2003, when a severe drought period occurred (Ciais et al., 2010) starting in mid-July and was considered by the calculation of the SPEI in the vulnerability analysis. The harvest of winter wheat at lower elevation started before the drought period, while the primary production at higher elevation was affected by the drought. The extension of the growing season allowed an over-compensation of NPP by a growing season that was 53 days longer, while the yield values were affected by drought, which could not be compensated for by a 7 day longer grain filling period. Both day of anthesis and day of maturity are determined based on temperature sums by the phenological model. For warmer areas, both will be earlier in the year compared to colder areas during the same period. In contrast, biomass production benefits from an extension of the growing season, due to additional days of production. However, the period between anthesis and maturity (both dates are represented by averages overall years and grid cells per resolution) is up to 5 days shorter when coarser resolution data are used for all models (Table 3). The models with the higher NPP at the higher elevation show the largest scale effect, which reflects a sensitive reaction of the phenological models to temperature changes. van Bussel et al. (2011) reported minimal scale effects on the phenological stages, but these changes have larger impacts on NPP.

4.2. Model specific aggregation effect

The differences between the resolutions of the wheat NPP simulations show three groups of models. APSIM, LINTUL, APSIMmod and AgroC show stronger effects (2.4–7.8%) than the other models, and HERMES, CENTURY, DailyDayCent and MONICA show minimal impacts (<1.2%), while COUP and STICS lie between (1.8–2.2%). The groups of models with medium and strong aggregation effects are all models that determine NPP based on the radiation use efficiency, while the other models use temperature based approaches. Hoffmann et al. (2015) investigated the aggregation effect on yields for the same set of models and the same region and suggested that the aggregation effect on radiation may not be much higher than on temperature, but the models might be sensitive to changes on radiation values. Despite trends of decreasing temperature and solar radiation at coarser resolution (Fig. 2 and Table 2 in Hoffmann et al., 2015), some models show increasing potential NPP values (LINTUL, APSIMmod and AgroC for wheat and DailyDayCent, LandscapeDNDC and AgroC for maize). These contradictory trends are also related to an extended growing season caused by different approaches for the calculation of the phenological stages. Therefore, decreasing temperatures affect an extension of the growing season which compensates, or over-compensates, for the effect of lower temperatures and radiation on crop growth as already discussed above. The growing season is extended by 1.5–5 days on average at coarser resolutions for wheat, and 1–2 days on average for maize simulations (Table 3), accompanied by a temperature decrease of 0.3 °C (Hoffmann et al., 2015). As model structure differs, the models show a different sensitivity to this effect, and differ in their trends through the different resolutions. These results concur with the findings by van Bussel et al. (2011), who found only small impacts of climate aggregation on determined phenological stages. The models APSIMmod, LINTUL and AgroC showed the longest (4–5 days) extension to the average growing season towards the coarser resolutions, while all other models showed lower extensions (<2.5 days). The models with the larger aggregation effects are the same models that show large differences in the length of the growing season.

The findings of the overall aggregation effect on NPP agrees with the findings of Nungesser et al. (1999), who found a mean uncertainty of <2% on NPP by modifying precipitation and solar radiation for resolutions of 10 and 50 km. The average differences for the wheat NN approach in this study (1.2%) is also in this range, but includes a large increase for AgroC of up to 3.3%. Pierce and Running (1995) determined an aggregation error of 15–30%, depending on the time step, while the maximum time period is one year and the higher values are related to impacts on daily NPP. The relatively small aggregation effect also agrees with the findings of de Wit et al. (2005), who found only small impacts of climate data aggregation on yield. Hoffmann et al. (2015) analysed the aggregation effect for wheat yield and found an uncertainty of 0.2 t ha⁻¹ (30 year average, over the entire study area and considering the simulation results of 13 models), which is about 2.5% of the yield. A comparison with the maximum aggregation effects on yield for each model shows 4.2–8.4% and 3.6–7.6% for wheat PN and WN, respectively, higher impacts on yield than for NPP, but also only small changes of the aggregation effect between the PN and WN simulations. Despite the high differences in precipitation between the five resolutions, the aggregation effect does not change strongly for the approaches WN and NN. For most models the aggregation effect even decrease. This can be explained by the limited relevance of drought and nitrogen stress on long term averages in the study region. In regions with higher drought or nutrient stress the aggregation effect might be greater. The region studied here was selected to represent an agricultural area in central Europe and because all necessary data were available at high resolution (1 km). The methodology can be

transferred to other comparable regions, but for regions of other climate conditions (higher drought stress) or with different management practices (higher nutrient stress) different aggregation effects might be observed, so further research needs to be done to address this.

The differences between the aggregation effect for the average and the median NPP reflects the impacts on the variance. Changes due to aggregation may be more relevant for more extreme years and show stronger impacts that are averaged out in the average values. It can be assumed that the extremes are more likely to impact negatively on the NPP values, since the maximum growth will be restricted. The strongest differences arising from the aggregation effect on averages and median are detected for models that show a decreasing aggregation effect towards the coarser resolution. The effect may be compensated by the general trend of the NPP values for coarser resolutions.

If the simulation runs are used as ensembles, the aggregation effect is relatively low. For the wheat simulation, the aggregation effect is almost as low as the minimal effect on an individual model run and can be explained by the different trends of the individual simulation approaches of the different models. In contrast, the simulations for maize show similar trends from the high resolution to the low resolution for models that show a high aggregation effect, which is reflected in a higher aggregation effect than for the wheat simulation. However, with difference of only 2%, this effect is still small.

4.3. Aggregation effect over different time periods

The analysis considered 29 year averages, while the impact of scale increases for shorter time periods (Figs. 6 and 7). The scale effect shows the highest differences for single years, and stabilises to constant differences or only minor changes for periods of 15 years and longer. This suggests that the resolution of choice depends on the temporal scale as well as on the research question. The mean uncertainty for longer periods will be below 4%. The maximum aggregation effect for single years can be more than 9% for wheat and more than 13% for maize, but will decrease to below 4% for time periods longer than 10 years. While the largest impact on the aggregation for the maximum differences is from 1 to 5 years, the mean aggregation effect shows the maximum change for the step from 25 to 29 years. NPP is a non-linear process and is especially affected by extreme events (Reichstein et al., 2013). Extreme weather conditions have an impact for a short period, often affecting only one year, and can be reduced by spatial averaging of the climate data. This spatial averaging is represented by the higher aggregation effect for the annual data, but this effect is already compensated by averaging over longer time periods. The impact of the temporal averages is represented in the two graphs in Fig. 7, which shows lower impacts on the averaged, short-term aggregation effect than on the maximum impact for a year.

Pierce and Running (1995) also found changes in response with aggregation depended on temporal scale. They observed decreasing aggregation errors for increasing periods, which supports our findings. However, their study considers daily to annual time periods and not periods lasting 1–29 years. An error of 15% is in agreement with the maximum values of the analysis of the annual aggregation effect, which shows a range of 1.6–9.4% for wheat and 3.4–13.6% for maize.

4.4. Vulnerability analysis

The impact of the simulated phenological stages has already been discussed in context of other effects, but is also important in the vulnerability analysis through varying the length of the growing season. Van Oijen et al., 2014 suggested the period from April to

September as the best period to determine the drought index, and found minimal impacts by starting this calculation of the drought index earlier, while we show a strong impact in this study. Van Oijen et al. (2014) used a biogeochemical model with fixed harvest dates for crop simulations or considered simulation results of forests and croplands with a fixed length of the growing season, but the models in this study are mainly crop models with dynamic growing season length. Therefore, in some years, the growing season ends before any drought impacts on crop growth occur, as for the year 2003. van Bussel et al. (2011) described the scale impact on the modelling of phenological stages as minor, but these changes can affect other processes as, in this case, the impact of a drought period on primary production. In this study, the relevant period to define hazardous conditions for the vulnerability analysis is from February to July, which is similar to the length of the growing season for most, but not all, models.

Not all models show sensitivity to the hazardous conditions defined by the SPEI. For the model AgroC (wheat simulations) and LandscapeDNDC (maize simulations), the threshold of the SPEI = −1 is not significant in comparison to model internal thresholds for drought stress. As the index SPEI is calculated by precipitation and temperature, there might be a discrepancy in the detection of extreme weather events by the APSIM models, which uses the radiation use efficiency to determine NPP. In contrast to the vulnerability analysis for wheat, the analysis for maize shows no impact of drought or negative values for vulnerability, with the exceptions STICS and MONICA, because maize is a C₄ plant and is more drought tolerant (Lopes et al., 2011). The comparison of the WN and PN simulation runs support the findings of the vulnerability analysis, as most models show only small differences. Analysis for higher thresholds is not useful for the considered study area, because the number of grid cells for hazardous conditions becomes too small for solid statistics. Regarding the resolution, the results show either all positive or negative values (HERMES, STICS, CENTURY, APSIMmod, LandscapeDNDC, APSIM, AgroC), or marginal differences to zero vulnerability (DailyDayCent, LINTUL, MONICA). The number of extreme weather events is expected to decrease at coarser scales, but the overall averages show little change between the resolutions. The expected pattern of stronger impacts of extreme events for finer scales with lower impacts at coarser scales is not seen. Assuming a threshold at SPEI = −1 to define drought conditions affects the same ratio of hazardous grid cells for the different resolutions. In contrast to the initial assumption, that with changes in the resolution will affect changes in the number of less extreme events, the number of grid-cells defined as extreme show the same or even higher ratio for the coarser resolution. One reason is the temporal scale for the NPP calculation, which is annual. In the vulnerability analysis the extreme events are defined during the first half of the year, which includes the drought of 2003 that had negative impacts on cropland NPP (Ciais et al., 2010), but not all model results are affected by this drought, because of an earlier harvest. Extreme events, therefore, appear to play a minor role in this area for long term averages.

The aggregation effect is stronger for the vulnerability than for the NPP averages. Apart for AgroC, and for the wheat simulation results of APSIMmod, the aggregation effect for the vulnerability at least doubles in comparison to the long term averages. The effect might be influenced by fewer years considered in the vulnerability analysis, but the maximum values even exceed the effect for annual averages. This shows that periods with extreme weather conditions are more strongly affected by the aggregation of climate input data than other years. When averaged over the long term, these impacts may even out.

5. Conclusions

NPP differs by up to around 8% and 5% for wheat and maize, respectively, depending on the spatial resolution of the climate input data, which is a slightly smaller aggregation effect than on yield. For most models, the overall averages are affected by only 2% or less, but for shorter time periods (shorter than 15 years), the aggregation effect may rise for annual NPP to over 9 or over 13% for wheat and maize, respectively. It is concluded that a large part of the aggregation effect is related to changes in phenology, which changes as result of a temperature decline from the finer to the coarser resolution. Since the simulated NPP averages show large differences between the single models and the aggregation effect can be high, ensemble runs can reduce the aggregation effect and address the problem of varying averages. The aggregation effect has a larger impact on the vulnerability analysis, which reflects the greater impact of aggregation effects for years with extreme weather conditions. A finer spatial resolution of climate input data will not greatly improve simulations for long term averages of NPP or vulnerability, but might reduce the aggregation effect for shorter periods. The greatest changes are detected for the steps from 1 km to 10 km resolution and from 50 km to 100 km. The current study suggests that long term NPP averages over large areas (e.g. regional scale) are relatively insensitive to climate data aggregation, whereas data aggregation could influence average NPP under extreme weather conditions. Based on these results, there is no need to simulate long term NPP averages at high resolution, if soil type and management do not vary in time and space. As the aggregation effect is not only limited to the aggregation of climate data, more work is required to investigate the impacts of heterogeneous soil types and varying management conditions.

Acknowledgement

This work was supported in part by the UK BBSRC through the “Modelling European Agriculture with Climate Change for Food Security” (MACSUR) project (BB/N004922/1).

References

- Bandaru, V., West, T.O., Ricciuto, D.M., Izaurrealde, R.C., 2013. Estimating crop net primary production using national inventory data and MODIS-derived parameters. *ISPRS J. Photogramm. Remote Sens.* 80, 61–71.
- Bergez, J.E., Chabrier, P., Gary, C., Jeuffroy, M.H., Makowski, D., Quesnel, G., Ramat, E., Raynal, H., Rousse, N., Wallach, D., Debaeke, P., Durand, P., Duru, M., Dury, J., Faverdin, P., Gascuel-Oudoux, C., Garcia, F., 2013. An open platform to build, evaluate and simulate integrated models of farming and agro-ecosystems. *Environ. Modell. Softw.* 39, 39–49.
- Brisson, N., Mary, B., Ripoche, D., Jeuffroy, M.H., Ruget, F., Nicoulaud, B., Gate, P., Devienne-barret, F., Recous, S., Tayot, X., Plenet, D., Cellier, P., Machet, J., Marc, J., Delécolle, R., 1998. STICS: a generic model for the simulation of crops and their water and nitrogen balances. 1. Theory and parameterization applied to wheat and corn. *Agronomie* 18, 311–346.
- Brisson, N., Launay, M., Mary, B., Beaudoin, N., 2008. Conceptual basis, formalisations and parameterization of the STICS crop model. Quae. ed.
- Cale, W.G., Oneill, R.V., Gardner, R.H., 1983. Aggregation error in non-linear ecological models. *J. Theor. Biol.* 100 (3), 539–550.
- Chen, C., Wang, E., Yu, Q., 2010. Modeling wheat and maize productivity as affected by climate variation and irrigation supply in North China Plain. *Agron. J.* 102, 1037–1049.
- Ciais, P., Wattenbach, M., Vuichard, N., Smith, P., Piao, S.L., Don, A., Luyssaert, S., Janssen, I.A., Bondeau, A., Dechow, R., Leip, A., Smith, P.C., Beer, C., van der Werf, G.R., Gervois, S., van Oost, K., Tomelleri, E., Freibauer, A., Schulze, E.D., CarboEurope synthesis team, 2010. The European carbon balance. Part 2: croplands. *Global Change Biol.* 16, 1409–1428. <http://dx.doi.org/10.1111/j.1365-2486.2009.02055.x>.
- Conrad, Y., Fohrer, N., 2009. Modelling of nitrogen leaching under complex winter wheat and red clover crop rotation on a drained agricultural field. *Phys. Chem. Earth* 34, 530–540.
- de Wit, A.J.W., Boogaard, H.L., van Diepen, C.A., 2005. Spatial resolution of precipitation and radiation: the effect on regional crop yield forecasts. *Agric. For. Meteorol.* 135, 156–168.
- Del Grosso, S.J., Parton, W.J., Mosier, A.R., Hartman, M.D., Brenner, J., Ojima, D.S., Schimel, D.S., 2001. Simulated interaction of carbon dynamics and nitrogen

- trace gas fluxes using the DAYCENT model. In: Schaffer, M. (Ed.), *Modeling Carbon and Nitrogen Dynamics for Soil Management*. CRC Press, Boca Raton, Florida, USA, pp. 303–332.
- Del Grosso, S.J., Parton, W., Mosier, A.R., Walsh, M.K., Ojima, D., Thornton, P.E., 2006. DAYCENT national scale simulations of N₂O emissions from cropped soils in the USA. *J. Environ. Qual.* 35, 1451–1460.
- Ewert, F., van Bussel, L.G.J., Zhao, G., Hoffmann, H., Gaiser, T., Specka, X., Nendel, C., Kersebaum, K.-C., Sosa, C., Lewan, E., Yeluripati, J., Kuhnert, M., Tao, F., Rötter, R., Constantin, J., Raynal, H., Wallach, D., Teixeira, E., Grosz, B., Bach, M., Doro, L., Roggero, P.P., Zhao, Z., Wang, E., Kiese, R., Haas, E., Eckersten, H., Trombi, G., Bindu, M., Klein, C., Biernath, C., Heinlein, F., Priesack, E., Cammarano, D., Asseng, S., Elliott, J., Glotter, M., Basso, B., Baigorria, G.A., Romero, C.C., Moriondo, M., 2015. *Uncertainties in scaling-up crop models for large-area climate change impact assessments*. In: Rosenzweig, C., Hillel, D. (Eds.), *Handbook of Climate Change and Agroecosystems*. ICP Series on Climate Change Impacts, Adaptation, and Mitigation, vol. 3. World Scientific Publishing Company, pp. 262–277.
- Friend, A.D., Lucht, W., Rademacher, T.T., Keribin, R., Betts, R., Patricia Cadule, P., Philippe Ciais, P., Douglas Clark, B.D.B., Dankers, R., Falloon, P.D., Ito, A., Kahana, R., Kleidon, A., Lomas, M.R., Nishina, K., Ostberg, S., Pavlick, R., Peylin, P., Schaphoff, S., Vuichard, N., Warszawski, L., Wiltshire, A., Woodward, F.I., 2014. Carbon residence time dominates uncertainty in terrestrial vegetation responses to future climate and atmospheric CO₂. *PNAS* 111 (9), 3280–3285.
- Gaiser, T., Perkins, U., Küpper, P.M., Kautz, T., Uteu-Puschmann, D., Ewert, F., Enders, A., Krauss, G., 2013. Modeling biopore effects on root growth and biomass production on soils with pronounced sub-soil clay accumulation. *Ecol. Modell.* 256, 6–15.
- Haas, E., Klatt, S., Fröhlich, A., Werner, C., Kiese, R., Grote, R., Butterbach-Bahl, K., 2012. LandscapedNDC: a process model for simulation of biosphere-atmosphere-hydrosphere exchange processes at site and regional scale. *Landsc. Ecol.*, <http://dx.doi.org/10.1007/s10980-012-9772-x>.
- Hemming, D., Betts, R., Collins, M., 2013. Sensitivity and uncertainty of modelled terrestrial net primary productivity to doubled CO₂ and associated climate change for a relatively large perturbed physics ensemble. *Agric. For. Meteorol.* 170, 79–88.
- Herbst, M., Hellebrand, H.J., Bauer, J., Huisman, J.A., Šimunek, J., Weihermüller, L., Graf, A., Vanderborght, J., Vereecken, H., 2008. *Multiyear heterotrophic soil respiration: evaluation of a coupled CO₂ transport and carbon turnover model*. *Ecol. Modell.* 214, 271–283.
- Heuvelink, G.B.M., Pebesma, E.J., 1999. Spatial aggregation and soil process modelling. *Geoderma* 89, 47–65.
- Hoffmann, H., Zhao, G., van Bussel, L.G.J., Enders, A., Specka, X., Sosa, C., Yeluripati, J., Tao, F., Constantin, J., Raynal, H., Teixeira, E.B., Grosz, B.L., Doro, L., Zhao, Z., Wang, E., Nendel, C., Kersebaum, K.C., Haas, E., Kiese, R., Klatt, S., Eckersten, H., Vanuytrec, D., Kuhnert, M., Lewan, E., Rötter, R., Roggero, P.P., Wallach, D., Cammarano, D., Asseng, S., Krauss, G., Siebert, S., Gaiser, T., Ewert, F., 2015. *Variability of effects of spatial climate data aggregation on regional yield simulation by crop models*. *Clim. Res.* 65, 53–69.
- Holzworth, D.P., Huth, N.I., deVoil, P.G., Zurcher, E.J., Herrmann, N.I., McLean, G., Chenu, K., van Oosterom, E.J., Snow, V., Murphy, C., Moore, A.D., Brown, H., Whish, J.P.M., Verrall, S., Fainges, J., Bell, L.W., Peake, A.S., Poulton, P.L., Hochman, Z., Thorburn, P.J., Gaydon, D.S., Dalglish, N.P., Rodriguez, D., Cox, H., Chapman, S., Doherty, A., Teixeira, E., Sharp, J., Cichota, R., Vogeler, I., Li, F.Y., Wang, E., Hammer, G.L., Robertson, M.J., Dimes, J.P., Whitbread, A.M., Hunt, J., van Rees, H., McClelland, T., Carberry, P.S., Hargreaves, J.N.G., MacLeod, N., McDonald, C., Harsdorf, J., Wedgwood, S., Keating, B.A., 2014. *APSIM—evolution towards a new generation of agricultural systems simulation*. *Environ. Modell. Softw.* 62, 327–350.
- Liu, C., Dong, X., Liu, Y., 2015. Changes of NPP and their relationship to climate factors based on the transformation of different scales in Gansu, China. *Catena* 125, 190–199.
- Jansson, P.E., Karlberg, L., 2004. Coupled Heat and Mass Transfer Model for Soil-plant-atmosphere Systems. Royal Institute of Technology, Department of Civil and Environmental Engineering, Stockholm, Sweden, pp. 435. <http://www2.lwr.kth.se/Vara%20Datorprogram/CoupModel/NetHelp/default.htm> (last accessed 10.06.14).
- Keating, B.A., Carberry, P.S., Hammer, G.L., Probert, M.E., Robertson, M.J., Holzworth, D.P., Huth, N.I., Hargreaves, G., Meinke, H., Hochman, Z., Maclean, G., Verburg, K., Snow, V., Dimes, J.P., Silburn, M., Wang, E., Brown, S., Bristow, K.L., Asseng, S., Chapman, S.C., McCown, R.L., Freebairn, D.M., Smith, C.J., 2003. An overview of APSIM, a model designed for farming systems simulation. *Eur. J. Agron.* 18, 267–288.
- Kersebaum, K.C., 2007. Modelling nitrogen dynamics in soil-crop systems with HERMES. *Nutr. Cycl. Agroecosyst.* 77, 39–52.
- Kersebaum, K.C., 2011. Special features of the HERMES model and additional procedures for parameterization, calibration, validation, and applications. In: Ahuja, L.R., Ma, L. (Eds.), *Advances in Agriculture System Modeling Series 2*. American Society of Agronomy, Crop Science Society of America, Soil Science Society of America, pp. 65–94.
- Kraus, D., Weller, S., Klatt, S., Haas, E., Wassmann, R., Kiese, R., Butterbach-Bahl, K., 2015. A new LandscapedNDC biogeochemical module to predict CH₄ and N₂O emissions from lowland rice and upland cropping systems. *Plant Soil* 386 (1–2), 125–149.
- Lopes, M.S., Araus, J.L., van Heerden, P.D.R., Foyer, C.H., 2011. Enhancing drought tolerance in C4 crops. *J. Exp. Bot.* 62 (9), 3135–3153.
- Nendel, C., Berg, M., Kersebaum, K.C., Mirschel, W., Specka, X., Wegehenkel, M., Wenkel, K.O., Wieland, R., 2011. The MONICA model: testing predictability for crop growth, soil moisture and nitrogen dynamics. *Ecol. Modell.* 222, 1614–1625.
- Nungesser, M.K., Joyce, M.L.A., David, A., 1999. Effects of spatial aggregation on predictions of forest climate change response. *Cim. Res.* 11, 109–124.
- Pan, S., Tian, H., Dangal, S.R.S., Ouyang, Z., Tao, B., Ren, W., Lu, C., Running, S., 2014. Modeling and monitoring terrestrial primary production in a changing global environment: toward a multiscale synthesis of observation and simulation. *Adv. Meteorol.* 2014, 17, <http://dx.doi.org/10.1155/2014/965936> (Article ID 965936).
- Parton, W.J., Scurlock, J.M.O., Ojima, D.S., Gilmanov, T.G., Scholes, R.J., Schimel, D.S., Kirchner, T., Menaut, J.-C., Seastedt, T., Garcia Moya, E., Kamalrut, A., Kinyamario, J.L., 1993. Observations and modeling of biomass and soil organic matter dynamics for the grassland biome worldwide. *Glob. Biogeochem. Cycles* 7, 785–809.
- Parton, W.J., Scurlock, J.M.O., Ojima, D.S., Schimel, D.S., Hall, O., 1995. Impact of climate change on grassland production and soil carbon worldwide. *Glob. Change Biol.* 1, 13–22.
- Parton, W.J., Holland, E.A., Del Grosso, S.J., Hartman, M.D., Martin, R.E., Mosier, A.R., Ojima, D.S., Schimel, D.S., 2001. Generalized model for NO_x and N₂O emissions from soils. *J. Geophys. Res.* 106, 17403–17420.
- Pierce, L.L., Running, S.W., 1995. The effects of aggregating sub-grid land surface variation on large-scale estimates of net primary production. *Landsc. Ecol.* 10 (4), 239–253.
- Rastetter, E.B., King, A.W., Cosby, B.J., Hornberger, G.M., Oneill, R.V., Hobbie, J.E., 1992. Aggregating fine-scale ecological knowledge to model coarser-scale attributes of ecosystems. *Ecol. Appl.* 2, 55–70.
- Reich, P.B., Turner, D.P., Bolstad, P., 1999. An approach to spatially distributed modeling of net primary production (NPP) at the landscape scale and its application in validation of EOS NPP products. *Remote Sens. Environ.* 70, 69–81.
- Reichstein, M., Bahn, M., Ciais, P., Frank, D., Mahecha, M.D., Seneviratne, S.I., Zscheischler, J., Beer, C., Buchmann, N., Frank, D.C., Papale, D., Rammig, A., Smith, P., Thonicke, K., van der Velde, M., Vicca, S., Walz, A., Wattenbach, M., 2013. Climate extremes and the carbon cycle. *Nature* 500, 287–295.
- Schulze, E.D., Ciais, P., Luyssaert, S., Schrumpp, M., Janssen, I.A., Thiruchittampalam, B., Theloke, J., Saurat, M., Bringeau, S., Lelieveld, J., Lohila, A., Rebmann, C., Jung, M., Bastviken, D., Abril, G., Grass, G., Leip, A., Freibauer, A., Kutsch, W., Don, A., Nieschulze, J., Börner, A., Gash, J.H., Dolman, A.J., 2010. The European carbon balance. Part 4: integration of carbon and other trace-gas fluxes. *Glob. Change Biol.* 16, 1451–1469, <http://dx.doi.org/10.1111/j.1365-2486.2010.02215.x>.
- Shibu, M.E., Leffelaar, P.A., van Keulen, H., Aggarwal, P.K., 2010. LINTUL3, a simulation model for nitrogen-limited situations: application to rice. *Eur. J. Agron.* 32, 255–271.
- Thornton, C.W., 1948. An approach toward a rational classification of climate. *Geogr. Rev.* 38, 55–94.
- van Bussel, L.G.J., Ewert, F., Leffelaar, P.A., 2011. Effects of data aggregation on simulations of crop phenology. *Agric. Ecosyst. Environ.* 142, 75–84.
- Van Ittersum, M., Leffelaar, P., Van Keulen, H., Kropff, M., Bastiaans, L., Goudriaan, J., 2003. On approaches and applications of the Wageningen crop models. *Eur. J. Agron.* 18, 201–234.
- Van Oijen, M., Balkovi, J., Beer, C., Cameron, D.R., Ciais, P., Cramer, W., Kato, T., Kuhnert, M., Martin, R., Myneni, R., Rammig, A., Rolinski, S., Soussana, J.F., Thonicke, K., Van der Velde, M., Xu, L., 2014. Impact of droughts on the carbon cycle in European vegetation: a probabilistic risk analysis using six vegetation models. *Biogeosciences* 11, 6357–6375.
- Vicente-Serrano, S.M., Begueria, S., Lopez-Moreno, J.L., 2010. A multiscalar drought index sensitive to global warming: the standardized precipitation evapotranspiration index. *J. Clim.* 23, 1696–1718.
- Wang, E., Robertson, M., Hammer, G., Carberry, P., Holzworth, D., Meinke, H., Chapman, S., Hargreaves, J., Huth, N., McLean, G., 2002. Development of a generic crop model template in the cropping system model APSIM. *Eur. J. Agron.* 18, 121–140.
- Yeluripati, J.B., Van Oijen, M., Wattenbach, M., Neftel, A., Ammann, A., Parton, W.J., Smith, P., 2009. Bayesian calibration as a tool for initialising the carbon pools of dynamic soil models. *Soil Biol. Biochem.* 41, 2579–2583.
- Zaehle, S., Stich, S., Liski, J., Cramer, W., Erhard, M., Hickler, T., Smith, B., 2006. The importance of age-related decline in forest NPP for modelling regional carbon balances. *Ecol. Appl.* 16 (4), 1555–1574.
- Zhang, X., Izaurre, R.C., Manowitz, D.H., Sahajpal, R., West, T.O., Thomson, A.M., Xu, M., Zhao, K., LeDuc, S.D., Williams, J.R., 2015. Regional scale cropland carbon budgets: evaluating a geospatial agricultural modeling system using inventory data. *Environ. Modell. Softw.* 63, 199–216.
- Zhao, G., Hoffmann, H., Van Bussel, L.G.J., Enders, A., Specka, X., Sosa, C., Yeluripati, J., Tao, F.L., Constantin, J., Teixeira, E., Grosz, B., Doro, L., Zhao, Z., Nendel, C., Kiese, R., Raynal, H., Eckersten, H., Haas, E., Wang, E., Kuhnert, M., Trombi, G., Bindu, M., Lewan, E., Bach, M., Kersebaum, K.C., Rötter, R., Roggero, P.P., Wallach, D., Krauss, G., Siebert, S., Gaiser, T., Ewert, F., 2015. Effect of weather data aggregation on regional crop simulation for different crops, production conditions, and response variables. *Clim. Res.* 65, 141–157, <http://dx.doi.org/10.3354/cr01301>.